

Belirli İntegraller için Ortalama Değer Teoremi

Eğer f , $[a, b]$ de sürekli ise $[a, b]$ aralığındaki bir c noktasında aşağıdaki eşitlik doğrudur:

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx$$

$$\text{Ort değer} = \frac{1}{b-a} \int_a^b f(x) dx$$

İntegral Hesabın Temel Teoremi

I. Kısım: Eğer f fonksiyonu $[a, b]$ üzerinde sürekli ise bu durumda $F(x) = \int_a^x f(t) dt$ de $[a, b]$ üzerinde sürekli,

(a, b) de türevlenebilirdir ve türevi $f(x)$ dir:

$$F'(x) = \frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x)$$

Leibnitz Kuralı: (İntegral İşareti Altında Türev)

f sürekli ve $u(x)$ ile $v(x)$ türevlenebilen fonksiyonlar ise:

$$\frac{d}{dx} \left[\int_{u(x)}^{v(x)} f(t) dt \right] = f(v(x)) \cdot v'(x) - f(u(x)) \cdot u'(x) \text{ dir.}$$

$$\textcircled{*} f(x) = \int_x^3 e^{-t^2} dt \Rightarrow f'(x) = ?$$

$$e^{-9} \cdot 0 - e^{-x^2} \cdot 1 = \boxed{-e^{-x^2}}$$

$$f'(x) = e^{-3^2} \cdot (3)' - e^{-x^2} \cdot 1 = -e^{-x^2}$$

~~2x + x^2~~ + x^2 Leibniz

$$(x+3)(2-x) - x(5-x)$$

$$2x + 6 - 3x - 5x + x^2$$

$$6 - 6x = 0$$

$$x = 1$$

⊗ $G(x) = x^2 \int_0^{5x} e^{-t^2} dt \Rightarrow G'(0) = ?$ 0

0,,

$$G'(x) = 2x \cdot \int_0^{5x} e^{-t^2} dt + x^2 [5 \cdot e^{-(5x)^2} - 0]$$

$$G'(0) = 0$$

⊗ $f(x) = \int_x^{x+3} (1.5-t) dt$ integralini maksimum yapan x değeri?

$$f'(x) = 1.5 - (x+3) = 0$$

$$= 6 - 6x = 0 \Rightarrow x = 1 \text{ K.N}$$

$$f''(x) = -6 < 0 \quad x = 1 \text{ max.}$$

yapan x

⊗ $f(x) = \int_{\cos x}^{\sin x} \frac{1}{1+t^2} dt \Rightarrow f'(\frac{\pi}{4}) = ?$

$\frac{1}{\cos x} \cdot \cos x + \frac{1}{\sin x} \cdot \sin x$

$$f'(x) = \cos x \cdot \frac{1}{1-\sin^2 x} + \sin x \cdot \frac{1}{1-\cos^2 x}$$

$$= \frac{1}{\cos x} + \frac{1}{\sin x}$$

$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

$$f'(\frac{\pi}{4}) = \frac{1}{\frac{\sqrt{2}}{2}} + \frac{1}{\frac{\sqrt{2}}{2}} = \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} = \frac{4}{\sqrt{2}}$$

II. Kısım (Hesaplama Teoremi)

Eğer f , $[a, b]$ deki her noktada sürekli ve F , f in $[a, b]$ deki herhangi bir ters türevi ise:

$$\int_a^b f(x) dx = F(b) - F(a) \text{ dir.}$$

İntegrasyon Teknikleri

① Yerine Koyma (Değişken Değiştirme) Kuralı

Eğer $u=g(x)$, değer kümesi I aralığı olan türevlenebilir bir fonk. ve f de I üzerinde sürekli ise:

$$\textcircled{1} \int f(g(x)) \cdot g'(x) dx = \int f(u) du \text{ dur.}$$

$$(u=g(x) \Rightarrow du=g'(x) dx)$$

$$\textcircled{2} \int_a^b f(g(x)) \cdot g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

$$(u=g(x) \Rightarrow du=g'(x) dx \quad \begin{array}{cc} \text{Eski sınır} & \text{Yeni sınır} \\ x=a & \rightarrow u=g(a) \\ x=b & \rightarrow u=g(b) \end{array})$$

$$\textcircled{*} \int (x^3+x)^5 \cdot (3x^2+1) dx = ? \quad \begin{array}{l} u^5 du \\ x^3+x=u \Rightarrow (3x^2+1) dx = du \end{array}$$

$$I = \int \frac{(x^3+x)^5}{u^5} \cdot \frac{du}{du} = \int u^5 du = \frac{u^6}{6} + C = \frac{(x^3+x)^6}{6} + C \checkmark$$

$$\textcircled{*} \int x^2 \sin x^3 dx = ? \quad \begin{array}{l} x^3=u \\ 3x^2 dx = du \\ x^2 dx = \frac{du}{3} \end{array} \quad x^3=u \Rightarrow 3x^2 dx = du \Rightarrow x^2 dx = \frac{du}{3}$$

$$\int \frac{\sin u}{\sin u} \cdot \frac{du}{3} = \frac{1}{3} \int \sin u du = -\frac{\cos u}{3} + C = -\frac{\cos x^3}{3} + C$$

$$\textcircled{*} \int_{-1}^1 3x^2 \sqrt{x^3+1} dx = \int_0^2 \frac{du}{u^{1/2}} = \frac{u^{3/2}}{3/2} \Big|_0^2 = \frac{2}{3} \cdot 2\sqrt{2} = \frac{4\sqrt{2}}{3}$$

$$\begin{array}{l} x^3+1=u \\ 3x^2 dx = du \\ x=1 \Rightarrow u=2 \\ x=-1 \Rightarrow u=0 \end{array}$$

$$\begin{array}{l} x^3+1=u^2 \\ 3x^2 dx = 2u du \end{array}$$

$$\frac{2(x^3+1)^{3/2}}{3} \Big|_{-1}^1$$

$$\frac{4\sqrt{2}}{3} - 0 = \frac{4\sqrt{2}}{3} \quad \textcircled{1}$$

$$\int_{-1}^1 2u^2 du = \frac{2u^3}{3} \Big|_{-1}^1$$

$$2x+1=u^2 \quad \frac{u^2-1}{2} \cdot u \quad \frac{1}{2} \int u^3 - u \, du \quad \frac{1}{2} \left(\frac{u^4}{4} - \frac{u^2}{2} \right) \quad \frac{(2x+1)^2}{8} - \frac{2x+1}{4}$$

$$\textcircled{*} \int x \sqrt{2x+1} \, dx = \int \frac{u-1}{2} \cdot \sqrt{u} \frac{du}{2} = \frac{1}{4} \int (u^{3/2} - u^{1/2}) \, du$$

$$\left(\begin{array}{l} 2x+1=u \\ 2dx=du \\ x=\frac{u-1}{2} \end{array} \right) \quad = \frac{1}{4} \left(\frac{u^{5/2}}{5/2} - \frac{u^{3/2}}{3/2} \right) + C$$

$$= \frac{1}{10} \cdot (2x+1)^{5/2} - \frac{1}{6} (2x+1)^{3/2} + C$$

$$\textcircled{*} \int_0^8 \frac{\cos \sqrt{x+1}}{\sqrt{x+1}} \, dx = \int_1^3 2 \cos u \, du = 2 \sin u \Big|_1^3 = 2 \sin 3 - 2 \sin 1$$

$$\sqrt{x+1}=u \quad \frac{dx}{2\sqrt{x+1}}=du \quad \sqrt{x+1}=u \quad 2 \cos u \, du$$

$$x=8 \Rightarrow u=3 \quad x=0 \Rightarrow u=1 \quad \frac{1}{2\sqrt{x+1}} dx = du \quad 2 \sin u \int_1^3 \quad \boxed{2 \sin 3 - 2 \sin 1}$$

$$\textcircled{*} \int \frac{2x}{\sqrt[3]{x^2+1}} \, dx = \int \frac{du}{\sqrt[3]{u}} = \int u^{-1/3} \, du = \frac{u^{2/3}}{2/3} + C = \frac{3}{2} (x^2+1)^{2/3} + C$$

$$x^2+1=u \rightarrow 2x \, dx = du \quad x^2+1=u^3 \quad 2x \, dx = 3u^2 \, du \quad \int 3u^2 \, du \quad \frac{3u^3}{2} + C$$

Tanx, Cotx, Secx, Cosecx integralleri

$$\textcircled{*} \int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx = - \int \frac{du}{u} = - \ln|u| + C = - \ln|\cos x| + C$$

$$\cos x = u \quad \downarrow \quad -\sin x \, dx = du \quad \cos x = u \quad \frac{\sin x}{\cos x} \quad \frac{du}{u} \quad -\ln|u| + C \quad \boxed{-\ln|\cos x| + C}$$

$$\textcircled{*} \int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx = \int \frac{du}{u} = \ln|u| + C = \ln|\sin x| + C$$

$$\sin x = u \Rightarrow \cos x \, dx = du \quad \ln|u| + C \quad \frac{1}{u} \, du$$

$$\textcircled{*} \int \sec x \, dx = \int \sec x \cdot \frac{(\sec x + \tan x)}{(\sec x + \tan x)} \, dx = \int \frac{\sec^2 x + \sec x \cdot \tan x}{\sec x + \tan x} \, dx$$

$$\frac{1}{\cos x} \, dx \quad \sec x + \tan x = u \quad \downarrow \quad (\sec x \tan x + \sec^2 x) \, dx = du \quad \left\{ \begin{array}{l} = \int \frac{du}{u} = \ln|u| + C \\ = \ln|\sec x + \tan x| + C \end{array} \right. \quad \textcircled{2}$$

$$\frac{\cos x}{\sin x} =$$

$$\frac{1}{u} \, du$$

$$\sin x = u$$

$$\cos x \, dx = du$$

$$-\ln|\csc x + \cot x| + C$$

$$\textcircled{*} \int \csc x \, dx = \int \csc x \cdot \frac{(\csc x + \cot x)}{\csc x + \cot x} \, dx = \int \frac{\csc^2 x + \cot x \csc x}{\csc x + \cot x} \, dx$$

$$\begin{aligned} \csc x + \cot x &= u \\ \downarrow \\ -(\csc x \cot x + \csc^2 x) \, dx &= du \end{aligned} \quad \left\{ \begin{aligned} &= -\int \frac{du}{u} = -\ln|u| + C \\ &= -\ln|\cot x + \csc x| + C \end{aligned} \right.$$

Sin²x ve Cos²x İntegralleri

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \text{ve} \quad \cos^2 x = \frac{1 + \cos 2x}{2} \quad \text{özdeşlikleri kullanarak}$$

$$\cos 2x = 1 - 2\sin^2 x$$

$$\frac{\cos 2x - 1}{-2} = \sin^2 x$$

$$\textcircled{*} \int \sin^2 x \, dx = \int \frac{1 - \cos 2x}{2} \, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

$$\boxed{\frac{x}{2} - \frac{\sin 2x}{4} + C}$$

$$\textcircled{*} \int \cos^2 x \, dx = \int \frac{1 + \cos 2x}{2} \, dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$$

$$\boxed{\frac{x}{2} + \frac{\sin 2x}{4} + C}$$

Simetrik fonksiyonların Belirli İntegrali

$f(x)$, $[-a, a]$ da sürekli olsun.

a) f , çift fonksiyon ise : $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$ dir.

(Çünkü $f(x)$ y-eksenine göre simetriktir.

$$\left. \begin{aligned} &\text{A} \quad \text{A} \\ &\int_{-a}^a f(x) \, dx = 2A \\ &= 2 \int_0^a f(x) \, dx \end{aligned} \right\}$$

b) f , tek fonk. ise : $\int_{-a}^a f(x) \, dx = 0$ dir.

(Çünkü $f(x)$ orjine göre simetriktir.

$$\left. \begin{aligned} &\text{A} \quad \text{A} \\ &\int_{-a}^a f(x) \, dx = A - A = 0 \end{aligned} \right\}$$

③

Ters Trigonometrik fonksiyonları veren integraller

$$\textcircled{*} \int \frac{dx}{\sqrt{1-x^2}} = \text{ArcSin} x + C$$

$$\textcircled{*} \int \frac{dx}{1+x^2} = \text{Arc Tan} x + C$$

$$\int \frac{1}{4 + \frac{x^2}{4}} dx = \frac{1}{4} \int \frac{1}{1 + \left(\frac{x}{2}\right)^2} dx \quad \frac{1}{4} \left(2 \text{arctan} \frac{x}{2} \right)$$

$$\frac{1}{2} \text{arctan} \frac{x}{2} + C$$

$$\textcircled{*} \int \frac{dx}{4+x^2} = \frac{1}{4} \int \frac{dx}{1+\left(\frac{x}{2}\right)^2} = \frac{1}{2} \int \frac{du}{1+u^2} = \frac{1}{2} \text{Arc Tan} u + C = \frac{1}{2} \text{Arc Tan} \frac{x}{2} + C$$

$$(u = \frac{x}{2} \Rightarrow du = \frac{dx}{2})$$

KURAL:

$$\textcircled{*} \int \frac{dx}{a^2+x^2} = \frac{1}{a} \text{Arc Tan} \frac{x}{a} + C$$

$$\frac{1}{a^2} \left(\text{arctan} \frac{x}{a} \right) + C$$

$$\frac{1}{a^2} \int \frac{dx}{1+\left(\frac{x}{a}\right)^2}$$

$$\textcircled{*} \int \frac{dx}{\sqrt{4-x^2}} = \int \frac{dx}{2\sqrt{1-\left(\frac{x}{2}\right)^2}} = \int \frac{du}{\sqrt{1-u^2}} = \text{Arc Sin} u + C = \text{Arc Sin} \frac{x}{2} + C$$

$$\frac{dx}{\sqrt{4(1-\frac{x^2}{4})}}$$

$$\frac{x}{2} = u \rightarrow \frac{dx}{2} = du$$

$$\frac{1}{2} \int \frac{dx}{\sqrt{1-\left(\frac{x}{2}\right)^2}} = \frac{1}{2} \left(2 \text{arcsin} \frac{x}{2} \right) + C$$

KURAL : $\textcircled{*} \int \frac{dx}{\sqrt{a^2-x^2}} = \text{Arc Sin} \frac{x}{a} + C$

$$\textcircled{*} \int \frac{dx}{x^2+3} = \frac{1}{\sqrt{3}} \text{Arc Tan} \frac{x}{\sqrt{3}} + C$$

$$\int \frac{dx}{3\left(1+\frac{x^2}{3}\right)} = \frac{1}{3} \int \frac{dx}{1+\left(\frac{x}{\sqrt{3}}\right)^2} = \frac{\sqrt{3}}{3} \text{Arc Tan} \frac{x}{\sqrt{3}}$$

$$\textcircled{*} \int \frac{dx}{\sqrt{5-x^2}} = \text{Arc Sin} \frac{x}{\sqrt{5}} + C$$

$$\frac{dx}{\sqrt{5}\sqrt{1-\frac{x^2}{5}}} = \frac{1}{\sqrt{5}} \int \frac{dx}{\sqrt{1-\left(\frac{x}{\sqrt{5}}\right)^2}} = \frac{1}{\sqrt{5}} \text{Arc Sin} \frac{x}{\sqrt{5}} + C$$

$$\textcircled{*} \int \frac{dx}{x^2+8} = \frac{1}{2\sqrt{2}} \text{Arc Tan} \frac{x}{2\sqrt{2}} + C$$

$$\frac{1}{2\sqrt{2}} \int \frac{dx}{1+\frac{x^2}{8}} = \frac{1}{2\sqrt{2}} \cdot 2\sqrt{2} \cdot \text{Arc Tan} \frac{x}{2\sqrt{2}}$$

$$\textcircled{*} \int \frac{dx}{\sqrt{16-x^2}} = \text{Arc Sin} \frac{x}{4} + C$$

$$\frac{dx}{\sqrt{16}\sqrt{1-\frac{x^2}{16}}} = \frac{1}{4} \int \frac{dx}{\sqrt{1-\left(\frac{x}{4}\right)^2}} = \frac{1}{4} \text{Arc Sin} \frac{x}{4} + C$$

④

② Kısmi İntegrasyon

$u(x)$ ve $v(x)$ türemlenebilir fonksiyonlar olsun.

$$\frac{d}{dx}(u.v) = v \cdot \frac{du}{dx} + u \cdot \frac{dv}{dx}$$

↓

$$\int \frac{d}{dx}(u.v) dx = \int v \frac{du}{dx} dx + \int u \frac{dv}{dx} dx$$

↓

$$u.v = \int v du + \int u dv \Rightarrow \boxed{\int u dv = u.v - \int v du} \rightarrow \text{Kısmi İntegrasyon Formülü}$$

* Belirli integralde :

$$\boxed{\int_a^b u dv = u.v \Big|_a^b - \int_a^b v du}$$

* u deme sıralaması :

L A P T Ü
 ↳ Üstel fonk.
 ↳ Trigonometrik fonk.
 ↳ Polinom
 ↳ Ters Trig. fonk.
 ↳ Logaritmik fonk.

$$\begin{array}{c} \text{Türev} \\ u \xrightarrow{\quad} du \\ \text{integral} \\ dv \xrightarrow{\quad} v \end{array}$$

$$\textcircled{*} \int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C$$

$$x = u \rightarrow dx = du$$

$$e^x dx = du \rightarrow e^x = v$$

$$\begin{array}{l} x = u \\ dx = du \\ e^x = v \end{array}$$

$$\boxed{x.e^x - e^x + C}$$

$$\textcircled{*} \int \ln x dx = x \ln x - \int dx = x \ln x - x + C$$

$$\begin{array}{l} \ln x = u \\ \downarrow \\ \frac{dx}{x} = du \\ x = v \end{array}$$

$$\begin{array}{l} \ln x = u \\ dx = du \\ x = v \end{array}$$

$$\ln x \cdot x - x + C$$

$$\textcircled{*} \int x^2 \sin x dx = -x^2 \cos x + 2 \int x \cos x dx = -x^2 \cos x + 2x \sin x - 2 \int \sin x dx$$

$$\begin{array}{l} x^2 = u \\ \downarrow \\ 2x dx = du \\ \sin x dx = dv \\ -\cos x = v \end{array} \quad \begin{array}{l} x = u \\ \downarrow \\ dx = du \\ \cos x dx = dv \\ \sin x = v \end{array}$$

$$= -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

$$x^2 = u \quad 2x dx = du \quad uv - \int v du$$

$$\begin{array}{l} \sin x dx = dv \\ -\cos x = v \end{array} \quad \begin{array}{l} \int \cos x \cdot x dx \\ -x^2 \cos x + 2x \sin x + 2 \cos x + C \end{array}$$

$$\begin{array}{l} x = u \\ \cos x dx = dv \\ \sin x = v \end{array} \quad \begin{array}{l} \int \sin x dx \\ uv - \int v du \\ + 2(x \sin x + \cos x) \end{array}$$

⑤

$$\arcsin x = u \quad \frac{1}{\sqrt{1-x^2}} dx = du$$

$$dx = dv \quad x = v$$

$$x \cdot \arcsin x + \sqrt{1-x^2} + C //$$

$$1-x^2 = a^2$$

$$-2x dx = 2a da$$

$$-x dx = a da$$

$$u \cdot v - \int v \cdot du$$

$$\int \frac{x}{\sqrt{1-x^2}} dx$$

$$\int \frac{1}{2} da$$

$$-1 \cdot a$$

$$-a$$

$$-\sqrt{1-x^2}$$

$$\cos x = u \quad -\sin x dx = du$$

$$e^x \cdot dx = dv \quad e^x = v$$

$$I = e^x \cos x + e^x \sin x - \int e^x \cos x dx$$

$$I = \frac{e^x (\cos x + \sin x)}{2}$$

$$\sin x = u \quad \cos x dx = dv$$

$$e^x \cdot dx = dv \quad e^x = v$$

$$e^x \sin x - \int e^x \cos x dx$$

$$e^x \cos x dx$$

$$\ln^2 x = u \quad 2 \ln x \cdot \frac{1}{x} dx = dv$$

$$x^3 dx = dv \quad \frac{x^4}{4} = v$$

$$\ln^2 x \cdot \frac{x^4}{4} \Big|_1^e - \frac{1}{2} \int_1^e x^3 \ln x dx$$

$$\frac{8e^4}{32} - \frac{-3e^4-1}{32}$$

$$\frac{5e^4-1}{32} //$$

$$\ln x = u \quad \frac{1}{x} dx = dv$$

$$x^3 dx = dv \quad \frac{x^4}{4} = v$$

$$-\frac{1}{2} \left(\ln x \cdot \frac{x^4}{4} \Big|_1^e - \frac{x^4}{16} \Big|_1^e \right)$$

$$-\frac{1}{2} \left(\frac{e^4}{4} - \left(\frac{e^4}{16} - \frac{1}{16} \right) \right)$$

$$-\frac{1}{2} \left(\frac{3e^4+1}{16} \right)$$

$$-\frac{3e^4-1}{32}$$

LAPTO

$$\arctan x = u \quad \frac{1}{1+x^2} dx = du$$

$$x dx = dv \quad \frac{x^2}{2} = v \quad \frac{1}{2} \int \frac{x^2}{1+x^2} dx$$

$$\frac{1}{2} \int 1 - \frac{1}{1+x^2} dx$$

$$\frac{x - \arctan x}{2}$$

$$\int x \arctan x dx = \frac{x^2}{2} \arctan x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx$$

$$\arctan x = u \quad \frac{x^2}{2} = v \quad x dx = dv$$

$$\frac{dx}{1+x^2} = du \quad \frac{x^2}{2} = v$$

$$= \frac{x^2}{2} \arctan x - \frac{1}{2} \left(1 - \frac{1}{1+x^2} \right) dx$$

$$= \frac{x^2}{2} \arctan x - \frac{x}{2} + \frac{\arctan x}{2} + C$$

LAPTO

$$\int \arcsin x dx = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx = x \arcsin x + \frac{1}{2} \int \frac{dt}{\sqrt{t}}$$

$$\arcsin x = u \quad dx = dv \quad \begin{cases} 1-x^2 = t \\ -2x dx = dt \end{cases}$$

$$\frac{dx}{\sqrt{1-x^2}} = dv \quad x = v$$

$$= x \arcsin x + \frac{1}{2} \cdot 2 \sqrt{1-x^2} + C$$

$$= x \arcsin x + \sqrt{1-x^2} + C$$

$$\int e^x \cos x dx = e^x \cos x + \int e^x \sin x dx$$

$$\cos x = u \quad e^x dx = dv \quad \begin{cases} \sin x = u \quad e^x dx = dv \\ \cos x dx = du \quad e^x = v \end{cases}$$

$$-\sin x dx = du \quad e^x = v$$

$$I = e^x \cos x + e^x \sin x - \int e^x \cos x dx$$

$$I$$

$$2I = e^x (\cos x + \sin x) \Rightarrow I = \frac{e^x}{2} (\cos x + \sin x) + C$$

$$\int_1^e x^3 \ln^2 x dx = \frac{x^4}{4} \ln^2 x \Big|_1^e - \frac{1}{2} \int_1^e x^3 \ln x dx = \frac{e^4}{4} - \frac{x^4}{4 \cdot 2} \ln x \Big|_1^e + \frac{1}{8} \int_1^e x^3 dx$$

$$\ln^2 x = u \quad x^3 dx = dv \quad \begin{cases} \ln x = u \quad x^3 dx = dv \\ \frac{2 \ln x}{x} dx = du \quad \frac{x^4}{4} = v \end{cases}$$

$$= \frac{e^4}{4} - \frac{e^4}{8} + \frac{x^4}{32} \Big|_1^e = \frac{5e^4-1}{32}$$

⑥

③ $\int \sin^m x \cdot \cos^n x dx$ integralleri

★ m çift, n tek ise: $u = \sin x$
 $du = \cos x dx$

★ n çift, m tek ise: $u = \cos x$
 $du = -\sin x dx$

dönüşümleri ve
 $\sin^2 x + \cos^2 x = 1$ özdeşliği
kullanılarak integral çözülür.

★ Hem m, hem de n tek ise herhangi birine u denir

★ Hem m, hem de n çift ise: $\cos^2 x = \frac{1 + \cos 2x}{2}$
 $\sin^2 x = \frac{1 - \cos 2x}{2}$ } özdeşlikleri ile
derece düşürülür

$$\textcircled{*} \int \sin^3 x \cdot \cos^8 x dx = \int \frac{\sin^2 x \cdot \cos^8 x \cdot \sin x dx}{1-u^2 \cdot u^8 \cdot -du} = \int u^8 (u^2 - 1) du$$

$$\begin{aligned} \cos x = u \quad -\sin x dx = du \\ \cos x = u \quad -\sin x dx = du \\ -\int \sin^2 x \cdot u^8 \cdot du &= -\int (1-u^2) \cdot u^8 du = -\int (u^8 - u^{10}) du \\ &= -\left(\frac{u^9}{9} - \frac{u^{11}}{11} \right) + C \\ &= \frac{(\cos x)^{11}}{11} - \frac{(\cos x)^9}{9} + C \end{aligned}$$

$$\textcircled{*} \int \cos^3(ax) dx = \int \frac{(\cos^2(ax))^2 \cdot \cos(ax) dx}{(1-u^2)^2 \cdot \frac{du}{a}} = \frac{1}{a} \int (1-2u^2+u^4) du$$

$$\begin{aligned} \sin ax = u \\ a \cos ax dx = du \\ = \frac{1}{a} \left(\sin x - \frac{2}{3} \sin^3 ax - \frac{1}{5} \sin^5 ax \right) + C \end{aligned}$$

$$\textcircled{*} \int \sin^5 x \cdot \cos^3 x dx = \int \frac{\sin^4 x \cdot \cos^2 x \cdot \cos x dx}{u^4 \cdot (1-u^2) \cdot du} = \int (u^5 - u^7) du = \frac{(\sin x)^6}{6} - \frac{(\sin x)^8}{8} + C$$

$$\sin x = u \quad \cos x dx = du$$

$$\textcircled{*} \int \sin^4 x dx = \int (\sin^2 x)^2 dx = \int \left(\frac{1 - \cos 2x}{2} \right)^2 dx = \frac{1}{4} \int (1 - 2\cos 2x + \cos^2 2x) dx$$

$$= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{1}{4} \int \frac{1 + \cos 4x}{2} dx$$

$$= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{x}{8} + \frac{\sin 4x}{32} + C$$

$$\begin{aligned} \int \cos^5(ax) dx &= \int \cos^4(ax) \cdot \cos(ax) dx \\ \sin ax = u \quad \cos ax dx = \frac{du}{a} \\ &= \frac{1}{a} \int (1-u^2)^2 \cdot u du = \frac{1}{a} \int (u - 2u^3 + u^5) du \\ &= \frac{1}{a} \left(\frac{u^2}{2} - \frac{2u^4}{4} + \frac{u^6}{6} \right) + C \\ &= \frac{1}{a} \left(\sin^2 ax - \frac{1}{2} \sin^4 ax + \frac{1}{6} \sin^6 ax \right) + C \end{aligned}$$

$$\begin{aligned} \int \sin^5 x \cdot \cos^3 x dx &= \int u^4 \cdot \cos^2 x \cdot du = \int u^4 (1-u^2) du \\ \sin x = u \quad \cos x dx = du \\ &= \int (u^4 - u^6) du = \frac{u^5}{5} - \frac{u^7}{7} + C \\ &= \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} + C \end{aligned}$$

$$\begin{aligned} \int \sin^4 x dx &= \int (\sin^2 x)^2 dx \\ \cos 2x = 1 - 2\sin^2 x \\ \sin^2 x = \frac{1 - \cos 2x}{2} \\ &= \frac{1}{4} \int (1 - 2\cos 2x + \cos^2 2x) dx \\ &= \frac{1}{4} \left(x - \sin 2x + \frac{\sin 4x}{4} \right) + C \\ &= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C \end{aligned}$$

$$\sec^2 x = 1 + \tan^2 x$$

$$\begin{aligned} \int \sec^4 x \, dx &= \int \sec^2 x \cdot \sec^2 x \, dx \\ \tan x &= u \\ \sec^2 x \, dx &= du \end{aligned} \quad \begin{aligned} &= \int (1+u^2) \, du \\ &= u + \frac{u^3}{3} + C = \tan x + \frac{\tan^3 x}{3} + C \end{aligned}$$

$$\begin{aligned} \int \sec^3 x \cdot \tan^3 x \, dx & \quad \begin{aligned} \sec^2 x &= 1 + \tan^2 x \\ \sec^2 x - 1 &= \tan^2 x \end{aligned} \\ \sec x &= u \\ \sec x \cdot \tan x \, dx &= du \end{aligned} \quad \begin{aligned} &\int \sec^2 x \cdot \tan^3 x \, dx \\ &\int u^2 (u^2 + 1) \, du \end{aligned} \quad \begin{aligned} &\int u^4 + u^2 \, du = \frac{u^5}{5} + \frac{u^3}{3} + C \\ &= \frac{\sec^5 x}{5} + \frac{\sec^3 x}{3} + C \end{aligned}$$

$$\begin{aligned} I &= \int \sec^3 x \, dx \\ \sec x &= u \quad \sec x \cdot \tan x \, dx = du \quad u \cdot u - \int u \cdot du \\ \sec^2 x \, dx &= du \quad \tan x = u \quad \sec x \cdot \tan x - \int \sec x \cdot \tan^2 x \, dx \\ 2I &= \sec x \cdot \tan x + \ln |\sec x + \tan x| - \int \sec^3 x - \sec x \, dx \\ I &= \frac{\sec x \cdot \tan x}{2} + \frac{\ln |\sec x + \tan x|}{2} + C \end{aligned}$$

④ $\int \sec^m x \cdot \tan^n x \, dx$ integralleri

- a) $u = \sec x \rightarrow du = \sec x \tan x \, dx$
b) $u = \tan x \rightarrow du = \sec^2 x \, dx$
c) Kısmi integrasyon

$\csc^2 x = 1 + \cot^2 x$
Yollarından biri ve $\sec^2 x = 1 + \tan^2 x$ özdeşliği kullanılarak çözülür.

$$\sec^2 x = 1 + \tan^2 x \quad \int \sec^2 x - 1 = \tan x - x + C$$

$$\textcircled{*} \int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx = \tan x - x + C$$

$$\textcircled{*} \int \sec^4 x \, dx = \int \frac{\sec^2 x}{1+u^2} \cdot \frac{\sec^2 x \, dx}{du} = \int (1+u^2) \, du = u + \frac{u^3}{3} + C$$

$$\begin{aligned} \tan x &= u \\ \downarrow \\ \sec^2 x \, dx &= du \end{aligned} \quad = \tan x + \frac{(\tan x)^3}{3} + C$$

$$\textcircled{*} \int \sec^5 x \cdot \tan^3 x \, dx = \int \frac{\sec^4 x}{u^2} \cdot \frac{\tan^2 x}{u^2-1} \cdot \frac{\sec x \cdot \tan x \, dx}{du} = \int (u^4 - u^2) \, du$$

$$\sec x = u \rightarrow \sec x \cdot \tan x \, dx = du \quad = \frac{\sec^5 x}{5} - \frac{\sec^3 x}{3} + C$$

$$\textcircled{*} I = \int \sec^3 x \, dx = \int \sec x \cdot \sec^2 x \, dx = \sec x \tan x - \int \sec x \cdot \frac{\tan^2 x \, dx}{\sec^2 x - 1}$$

$$\begin{aligned} \sec x &= u \quad \sec^2 x \, dx = du \\ \downarrow \quad \downarrow \\ \sec x \tan x \, dx &= du \quad v = \tan x \end{aligned} \quad = \sec x \tan x - \frac{\int \sec^3 x \, dx + \int \sec x \, dx}{I}$$

$$\Downarrow$$

$$2I = \sec x \tan x + \ln |\sec x + \tan x|$$

$$I = \frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + C$$

⑤ $\int \sin mx \cdot \cos nx \, dx$ İntegralleri

$\sin mx \cdot \cos nx$, $\sin mx \cdot \sin nx$, $\cos mx \cdot \cos nx$ çarpımlarını içeren integralle-
ri çözmek için:

$$\sin mx \cdot \cos nx = \frac{1}{2} (\sin(m-n)x + \sin(m+n)x)$$

$$\sin mx \cdot \sin nx = \frac{1}{2} (\cos(m-n)x - \cos(m+n)x)$$

$$\cos mx \cdot \cos nx = \frac{1}{2} (\cos(m-n)x + \cos(m+n)x)$$

dönüşümleri
kullanılır.

$$* \int \sin 3x \cdot \cos 5x \, dx = \frac{1}{2} \int (\sin(3x-5x) + \sin(3x+5x)) \, dx$$

$$\begin{aligned} & \int \frac{1}{2} (\sin(3x-5x) + \sin(3x+5x)) \, dx = \frac{1}{2} \int (-\sin 2x + \sin 8x) \, dx = \frac{\cos 2x}{4} - \frac{\cos 8x}{16} + C \\ & \frac{1}{2} \int \sin 8x - \sin 2x \, dx = \frac{1}{2} \left(-\frac{\cos 8x}{8} + \frac{\cos 2x}{2} \right) + C \end{aligned}$$

⑥ İndirgeme Formülleri



* $\int \tan^n x \, dx$ integrali için bir indirgeme formülü bulup,
bu formül yardımıyla $\int \tan^n x \, dx$ integralini hesaplayınız.

$$I_n = \int \tan^n x \, dx \text{ olsun.}$$

$$I_n = \int (\tan x)^{n-2} \cdot \tan^2 x \, dx = \int (\tan x)^{n-2} (\sec^2 x - 1) \, dx$$

$$= \int (\tan x)^{n-2} \cdot \sec^2 x \, dx - \int (\tan x)^{n-2} \, dx$$

(Tanx=u → Sec²x dx = du) I_{n-2}

$$= \int u^{n-2} \cdot du - I_{n-2} = \frac{u^{n-1}}{n-1} - I_{n-2}$$

$$I_n = \frac{(\tan x)^{n-1}}{n-1} - I_{n-2}$$

$$I_n = \frac{(\tan x)^{n-1}}{n-1} - I_{n-2}$$

$$\begin{aligned} I_5 &= \int (\tan x)^5 dx = \frac{(\tan x)^4}{4} - I_3 = \frac{(\tan x)^4}{4} - \left(\frac{(\tan x)^2}{2} - I_1 \right) \\ &= \frac{(\tan x)^4}{4} - \frac{(\tan x)^2}{2} + \int \tan x \\ &= \frac{(\tan x)^4}{4} - \frac{(\tan x)^2}{2} - \ln |\cos x| + C \end{aligned}$$

~~X~~ ~~★~~ zor

*) $\int x^n \cdot e^{ax} dx$ ($a \neq 0$) integrali için bir indirgeme formülü

bulup $\int x^3 \cdot e^{ax} dx$ integralini bu formül yardımı ile hesaplayın.

$$I_n = \int x^n e^{ax} dx \text{ olsun. } \begin{aligned} x^n &= u \rightarrow n \cdot x^{n-1} \cdot dx = du \quad x = \frac{du}{n} \\ e^{ax} dx &= dv \rightarrow \frac{e^{ax}}{a} = v \end{aligned}$$

$$I_n = \frac{1}{a} \cdot x^n \cdot e^{ax} - \frac{n}{a} \int x^{n-1} \cdot e^{ax} dx = \frac{1}{a} \cdot x^n \cdot e^{ax} - \frac{n}{a} I_{n-1}$$

$$I_n = \frac{1}{a} \cdot x^n \cdot e^{ax} - \frac{n}{a} I_{n-1}$$

$$I_3 = \int x^3 e^{ax} dx = \frac{1}{a} x^3 e^{ax} - \frac{3}{a} I_2 = \frac{1}{a} x^3 e^{ax} - \frac{3}{a} \left(\frac{1}{a} x^2 e^{ax} - \frac{2}{a} I_1 \right)$$

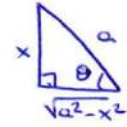
$$= \frac{1}{a} x^3 e^{ax} - \frac{3}{a^2} x^2 e^{ax} + \frac{6}{a^2} \left(\frac{1}{a} x e^{ax} - \frac{1}{a} I_0 \right)$$

$$= \frac{1}{a} x^3 e^{ax} - \frac{3}{a^2} x^2 e^{ax} + \frac{6}{a^3} x e^{ax} - \frac{6}{a^3} \int e^{ax} dx$$

$$= \frac{1}{a} x^3 e^{ax} - \frac{3}{a^2} x^2 e^{ax} + \frac{6}{a^3} x e^{ax} - \frac{6}{a^4} e^{ax} + C$$

⑦ Trigonometrik Değişken Dönüşümleri

$x = a \sin \theta$ $dx = a \cos \theta d\theta$



* $\theta = \text{Arc Sin } \frac{x}{a}$ dönüştürme denktir

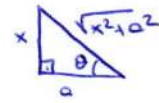
★ $\sqrt{a^2 - x^2}$ ($a > 0$) seklinde terim içeren integrallerde kullanılır.

* $\sqrt{a^2 - x^2}$ ifadesi $-a \leq x \leq a$ için anlamlıdır; bu ise $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ için mümkündür.

* $x = a \sin \theta$ dönüşümü ile:

$$\boxed{\begin{aligned} \sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} = a |\cos \theta| = a \cos \theta \\ x &= a \sin \theta \quad dx = a \cos \theta d\theta \end{aligned}}$$

⑥ $x = a \tan \theta$ dönüşümü

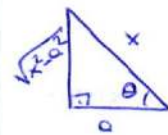


★ $\theta = \text{Arc Tan } \frac{x}{a}$ dönüşümüne denktir

★ $\sqrt{a^2+x^2}$ veya a^2+x^2 ($a>0$) içeren integrallerde kullanılır. $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ için:

$$\sqrt{a^2 + x^2} = \sqrt{a^2 + a^2 \tan^2 \theta} = a |\sec \theta| = a \sec \theta$$
$$x = a \tan \theta \rightarrow dx = a \sec^2 \theta d\theta$$

$x = a \sec \theta$ $\frac{dx}{d\theta} = a \sec \theta \tan \theta$



* $\theta = \text{Arc Sec } \frac{x}{a}$ dönüşümüne denktir

★ $\sqrt{x^2 - a^2}$ igeren integrallende koeffizienten

* ArcSec θ nın tanım kümesi: $0 \leq \theta \leq \frac{\pi}{2}$ ve $\frac{\pi}{2} < \theta \leq \pi$ dir.
işlem kolaylığı için $0 \leq \theta < \frac{\pi}{2}$ seçersek:

$$\sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2 \theta - a^2} = a |\tan \theta| = a \tan \theta$$

$$x = a \sec \theta \rightarrow dx = a \sec \theta \tan \theta d\theta$$

$$\int \frac{1}{\sqrt{5-x^2}} \cdot \frac{1}{5-x^2} dx = \int \frac{1}{\sqrt{5} \sqrt{1-\frac{x^2}{5}}} \cdot \frac{1}{5-5 \cdot \frac{x^2}{5}} dx = \int \frac{1}{\sqrt{5} \cos \theta} \cdot \frac{1}{5 \cos^2 \theta} = \frac{1}{5\sqrt{5}} \int \frac{1}{\cos^3 \theta} = \frac{1}{5\sqrt{5}} \int \sec^3 \theta d\theta$$

$$(*) \int \frac{dx}{(5-x^2)^{3/2}} = \int \frac{\sqrt{5} \cos \theta d\theta}{5\sqrt{5} \cos^3 \theta} = \frac{1}{5} \int \sec^2 \theta d\theta = \frac{1}{5} \tan \theta + C$$

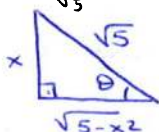
$$x = \sqrt{5} \sin \theta$$

$$dx = \sqrt{5} \cos \theta d\theta$$

$$\sqrt{5-x^2} = \sqrt{5(1-\sin^2 \theta)} = \sqrt{5} \cos \theta$$

$$\sin \theta = \frac{x}{\sqrt{5}} \quad \sqrt{5} \sin \theta = x$$

$$\sqrt{5} \cos \theta d\theta = dx$$

$$\frac{1}{5} \cdot \frac{x}{\sqrt{5-x^2}} + C$$


$$(*) \int \frac{x^2}{\sqrt{1-x^2}} dx = \int \frac{\sin^2 \theta}{\cos \theta} \cos \theta d\theta = \int \sin^2 \theta d\theta = \int \frac{1-\cos 2\theta}{2} d\theta$$

$$x = \sin \theta$$

$$dx = \cos \theta d\theta$$

$$\sqrt{1-x^2} = \cos \theta$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\sin \theta = x$$

$$\cos \theta d\theta = dx$$

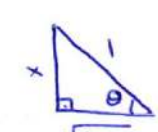
$$\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\frac{1-\cos 2\theta}{2}$$

$$\int \frac{1-\cos 2\theta}{2} d\theta = \frac{\theta}{2} - \frac{\sin 2\theta}{4} + C$$

$$= \frac{\theta}{2} - \frac{\sin \theta \cos \theta}{2} + C$$

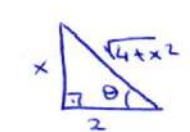
$$= \frac{\text{ArcSin } x}{2} - \frac{1}{2} x \sqrt{1-x^2} + C$$


$$(*) \int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2 \sec^2 \theta d\theta}{2 \sec \theta} = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C$$

$$x = 2 \tan \theta$$

$$dx = 2 \sec^2 \theta d\theta$$

$$\sqrt{4+x^2} = 2 \sec \theta$$

$$\ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + C$$


$$(*) \int \frac{dx}{(1+9x^2)^2} = \int \frac{1}{3} \cdot \frac{\sec^2 \theta d\theta}{\sec^4 \theta} = \frac{1}{3} \int \cos^2 \theta d\theta = \frac{1}{3} \int \frac{1+\cos 2\theta}{2} d\theta$$

$$3x = \tan \theta$$

$$3dx = \sec^2 \theta d\theta$$

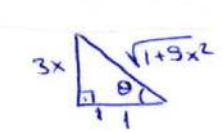
$$1+9x^2 = \sec^2 \theta$$

$$\frac{1}{6} \left(\frac{\sin 2\theta}{2} + \frac{\theta}{2} \right)$$

$$\frac{\sin 2\theta}{12} + \frac{\theta}{6}$$

$$\frac{\theta}{6}$$

$$\frac{\theta}{6} + \frac{1}{6} \sin \theta \cos \theta + C$$

$$= \frac{\text{ArcTan } x}{6} + \frac{1}{6} \cdot \frac{3x}{\sqrt{1+9x^2}} \cdot \frac{1}{\sqrt{1+9x^2}} + C$$


$$(*) \int \frac{dx}{x\sqrt{x^2-4}} = \int \frac{2 \sec \theta \tan \theta d\theta}{2 \sec \theta \cdot 2 \tan \theta} = \int \frac{d\theta}{2} = \frac{1}{2} \theta + C = \frac{1}{2} \text{ArcSec } \frac{x}{2} + C$$

$$x = 2 \sec \theta$$

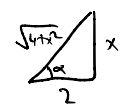
$$dx = 2 \sec \theta \tan \theta d\theta$$

$$\sqrt{x^2-4} = \sqrt{4(\sec^2 \theta - 1)} = 2 \tan \theta$$

$$-\frac{1}{2} \text{arccsc } \frac{x}{2} + C$$

(12)

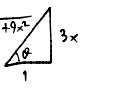
$$\int \frac{dx}{\sqrt{4+x^2}} = \int \frac{1}{2} \sec^2 \theta d\theta = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C$$

$$\ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + C$$


$$\int \frac{dx}{(1+9x^2)^2} = \int \frac{\sec^2 \theta d\theta}{3 \sec^4 \theta} = \frac{1}{3} \int \cos^2 \theta d\theta$$

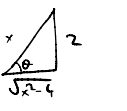
$$\frac{1}{3} \left(\frac{\sin 2\theta}{2} + \frac{\theta}{2} \right)$$

$$\frac{\sin 2\theta}{12} + \frac{\theta}{6}$$

$$\frac{\theta}{6}$$


$$\int \frac{dx}{x\sqrt{x^2-4}} = \int \frac{1}{2} \sec \theta \tan \theta d\theta = \int \frac{1}{2} d\theta = \frac{1}{2} \theta + C$$

$$\frac{1}{2} \theta + C$$

$$\theta = \text{arccsc } \frac{x}{2}$$


⑧ Rasyonel Fonksiyonların Basit Kesirlerle İntegrasyonu

Rasyonel fonksiyonların daha basit kesirlerin toplamı olarak yazma işlemine "Basit Kesirler Yöntemi" denir.

$$\frac{5x-3}{x^2-2x-3} = \frac{A}{x+1} + \frac{B}{x-3} \quad A, B \rightarrow \text{Belirsiz Katsayılar}$$

* $\frac{P(x)}{Q(x)}$ rasyonel fonksiyonunu basit kesirlere ayırabilmek için: $d(P(x)) < d(Q(x))$ olmalıdır. Değilse polinom bölmesi ile bu hale getirilir.

Basit Kesirlere Ayırma Yöntemi:

① $x-r$, $Q(x)$ in lineer (1. dereceden) çarpanı olsun.
 $(x-r)^m$, $Q(x)$ i bölen $(x-r)$ nin en büyük kuvveti olsun.
O zaman bu çarpan için m tane kısmi kesirin toplamı yazılır:

$$(x-r)^m \text{ çarpanı için: } \frac{A_1}{x-r} + \frac{A_2}{(x-r)^2} + \dots + \frac{A_m}{(x-r)^m}$$

Bu işlem, $Q(x)$ in her lineer çarpanı için yapılır.

② x^2+px+q , $Q(x)$ in kuadratik (2. dereceden) çarpanı olsun.
 $(x^2+px+q)^n$, bu çarpanın $Q(x)$ i bölen en büyük kuvveti olsun

$$(x^2+px+q)^n \text{ için: } \frac{B_1x+C_1}{x^2+px+q} + \frac{B_2x+C_2}{(x^2+px+q)^2} + \dots + \frac{B_nx+C_n}{(x^2+px+q)^n} \quad \text{Kısmi kesirleri yazılır.}$$

③ Orijinal $\frac{P(x)}{Q(x)}$ kesri, tüm bu basit kesirlerin toplamına

esitlenir. Paydalar eşitlenip, paylar da polinom eşitliği yardımıyla katsayılar hesaplanır.

* Kısmi (Heaviside) Yöntemi

$d(P(x)) < d(Q(x))$ ve $Q(x) = (x-r_1)(x-r_2)\dots(x-r_n)$, $Q(x)$ in birbirinden farklı, kuvveti 1 olan lineer çarpanları ise:

$$\frac{P(x)}{Q(x)} = \frac{P(x)}{(x-r_1)(x-r_2)\dots(x-r_n)} = \frac{A_1}{x-r_1} + \frac{A_2}{x-r_2} + \dots + \frac{A_n}{x-r_n} \quad \text{olur ve}$$

$$A_i = \lim_{x \rightarrow r_i} (x-r_i) \cdot \frac{P(x)}{(x-r_1)\dots(x-r_n)} \quad \text{dir.}$$

(Pratikte bunu, $\frac{P(x)}{Q(x)}$ in paydasındaki $x-r_i$ çarpanını çıkarıp kalan ifadeye $x=r_i$ yazarak yaparız.)

*) $\int \frac{5x-3}{x^2-2x-3} dx = ?$ $\frac{A}{x-3} + \frac{B}{x+1}$ $\int \frac{3}{x-3} + \frac{2}{x+1} dx = 3 \ln|x-3| + 2 \ln|x+1| + c$

$\frac{5x-3}{(x-3)(x+1)} = \frac{A}{x-3} + \frac{B}{x+1}$ $A = \frac{5x-3}{x+1} \Big|_{x=-1} = 2$ $B = \frac{5x-3}{x-3} \Big|_{x=3} = 3$

$A+B=5$ $A-3B=-3$ $4B=8$ $B=2$ $A=3$

$\int \frac{5x-3}{x^2-2x-3} dx = \int \left(\frac{3}{x-3} + \frac{2}{x+1} \right) dx = 3 \ln|x-3| + 2 \ln|x+1| + c$

$$\frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+3} = \frac{6}{8}$$

$$A = \frac{x^2+4x+1}{(x+1)(x+3)} \Big|_1 = \frac{3}{4}$$

$$B = \frac{x^2+4x+1}{(x-1)(x+3)} \Big|_{-1} = \frac{-2}{-4} = \frac{1}{2}$$

$$C = \frac{x^2+4x+1}{(x-1)(x+1)} \Big|_{-3} = \frac{-2}{+8} = -\frac{1}{4}$$

$$\frac{3}{4} \ln|x-1| + \frac{1}{2} \ln|x+1| - \frac{1}{4} \ln|x+3| + c$$

*) $I = \int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)} dx = ?$

$$\frac{x^2+4x+1}{(x-1)(x+1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+3}$$

$$I = \int \left(\frac{3}{4} \cdot \frac{1}{x-1} + \frac{1}{2} \cdot \frac{1}{x+1} + \frac{1}{4} \cdot \frac{1}{x+3} \right) dx$$

$$= \frac{3}{4} \ln|x-1| + \frac{1}{2} \ln|x+1| + \frac{1}{4} \ln|x+3| + c$$

$$A = \frac{x^2+4x+1}{(x+1)(x+3)} \Big|_{x=1} = \frac{6}{2 \cdot 4} = \frac{3}{4}$$

$$B = \frac{x^2+4x+1}{(x-1)(x+3)} \Big|_{x=-1} = \frac{-2}{-2 \cdot 2} = \frac{1}{2}$$

$$C = \frac{x^2+4x+1}{(x-1)(x+1)} \Big|_{x=-3} = \frac{9-12+1}{-4 \cdot -2} = -\frac{1}{4}$$

$$\frac{+9-12+1}{-4 \cdot -2} = \frac{-2}{8}$$

Çok güzel bir yöntem

$$\frac{A}{x+2} + \frac{B}{(x+2)^2} = \frac{Ax+2A+B}{(x+2)^2} = 6x+7 \quad A=6 \quad B=-5$$

$$\textcircled{*} \int \frac{6x+7}{(x+2)^2} dx = ?$$

$$\int \frac{6}{x+2} - \frac{5}{(x+2)^2} = 6 \ln|x+2| + \frac{5}{x+2} + C$$

$$\frac{6x+7}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2} = \frac{Ax+2A+B}{(x+2)^2}$$

$$2A+B=7 \Rightarrow B=-5$$

$$x^4 \frac{x^5}{5}$$

$$-5(x+2)^2$$

$$-5(x+2)^{-1}$$

$$-1$$

$$\frac{5}{x+2}$$

$$\int \frac{6x+7}{(x+2)^2} dx = \int \left(\frac{6}{x+2} - \frac{5}{(x+2)^2} \right) dx = 6 \ln|x+2| + \frac{5}{x+2} + C$$

$$\textcircled{*} \int \frac{x^3-4x^2-5x+3}{x^2-4x-5} dx = ?$$

$$\frac{x(x^2-4x-5)+x-3}{x^2-4x-5} = x + \frac{x-3}{(x-5)(x+1)}$$

$$\frac{x-3}{x^2-4x-5} = \frac{A}{x-5} + \frac{B}{x+1} \Rightarrow A = \frac{1}{3} \quad B = \frac{2}{3}$$

$$\frac{x(x^2-4x-5)}{x^2-4x-5} = x + \frac{x-3}{x^2-4x-5}$$

$$A = \frac{x-3}{x-5} \Big|_{x=5} = \frac{2}{3} \quad B = \frac{x-3}{x+1} \Big|_{x=-1} = \frac{4}{3}$$

$$\frac{x^2}{2} + \frac{1}{3} \ln|x-5| + \frac{2}{3} \ln|x+1| + C$$

$$I = \int \left(x + \frac{1}{3} \cdot \frac{1}{x-5} + \frac{2}{3} \cdot \frac{1}{x+1} \right) dx = \frac{x^2}{2} + \frac{1}{3} \ln|x-5| + \frac{2}{3} \ln|x+1| + C$$

$$\textcircled{*} I = \int \frac{4-2x}{(x^2+1)(x-1)} dx = ?$$

$$\frac{A}{x-1} + \frac{Bx+C}{x^2+1} = \frac{Ax^2+Ax+Cx-Bx-C}{(x-1)(x^2+1)} = 4-2x$$

$$\begin{aligned} A+B &= 0 & A &= -B \\ C-B &= -2 & & \\ A-C &= 4 & & \\ C+A &= 2 & & \end{aligned}$$

$$A+B=0$$

$$C-A=-2$$

$$B-C=4$$

$$B=1 \quad A=-1 \quad C=-3$$

$$\frac{4-2x}{(x^2+1)(x-1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1} = \frac{Ax^2-Ax+Bx^2+B+Cx-C}{(x-1)(x^2+1)}$$

$$\int \frac{1}{x-1} + \frac{-x-3}{x^2+1} dx = \int \frac{1}{x-1} - \int \frac{x}{x^2+1} - \int \frac{3}{x^2+1}$$

$$I = \int \left(\frac{x+3}{x^2+1} + \frac{1}{x-1} \right) dx = \int -\frac{x}{x^2+1} dx + \int \frac{3}{x^2+1} dx + \int \frac{dx}{x-1}$$

$$= -\frac{1}{2} \ln|x^2+1| + 3 \arctan x + \ln|x-1| + C$$

$$-x$$

$$\int \frac{x}{x^2+1} dx$$

$$x^2+1 = u$$

$$2x dx = du$$

$$\int \frac{1}{2u} du$$

$$-\frac{1}{2} \ln|x^2+1|$$

$$\frac{1}{2} \int \frac{1}{u} du$$

$$\frac{1}{2} \ln|u|$$

Yapı/mad

9) $\tan \frac{x}{2}$ Dönüşümü

$\sin x$ veya $\cos x$ içeren kesirli fonksiyonların integralini alırken kullanılır.

$\tan \frac{x}{2} = u$ dönüşümü ile: 

$$x = 2 \arctan u \Rightarrow dx = \frac{2}{1+u^2} du \text{ olur.}$$

$$\sin x = 2 \cdot \sin \frac{x}{2} \cdot \cos \frac{x}{2} = 2 \cdot \frac{u}{\sqrt{1+u^2}} \cdot \frac{1}{\sqrt{1+u^2}} = \frac{2u}{1+u^2}$$

$$\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} = \frac{1}{1+u^2} - \frac{u^2}{1+u^2} = \frac{1-u^2}{1+u^2}$$

Özet

$$u = \tan \frac{x}{2} \text{ ile:}$$

$$dx = \frac{2 du}{1+u^2}$$

$$\sin x = \frac{2u}{1+u^2}$$

$$\cos x = \frac{1-u^2}{1+u^2}$$

$$\textcircled{*} \int \frac{dx}{1+\cos x} = \int \frac{2 du}{1+\frac{1-u^2}{1+u^2}} = \int \frac{2}{\frac{1+u^2+1-u^2}{1+u^2}} du = \int \frac{2}{2} du = \int du = u + C = \tan \frac{x}{2} + C$$

$$\tan \frac{x}{2} = u \quad dx = \frac{2 du}{1+u^2}$$

$$\cos x = \frac{1-u^2}{1+u^2}$$

$$\textcircled{*} \int \frac{dx}{1-\sin x + \cos x} = \int \frac{2 du}{1-\frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}} = \int \frac{2 du}{\frac{1+u^2-2u+1-u^2}{1+u^2}} = \int \frac{2 du}{2-2u} = \int \frac{du}{1-u} = -\ln|1-u| + C = -\ln|1-\tan \frac{x}{2}| + C$$

$$\tan \frac{x}{2} = u \quad dx = \frac{2 du}{1+u^2}$$

$$1 - \sin x + \cos x = 1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2} = \frac{2(1-u)}{1+u^2}$$

$$\int \frac{dx}{1+\cos x}$$

$$\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}$$

$$\frac{1}{1+\cos x} = \frac{1}{1+\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}$$

$$\frac{2}{1+\cos x} = \frac{2}{1+\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}$$

$$\frac{1-u^2}{1+u^2} + 1 = \frac{1-u^2+1+u^2}{1+u^2} = \frac{2}{1+u^2}$$

$$u + C$$

$$\tan \frac{x}{2} + C$$

$$\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}$$

$$\frac{1}{1+u^2} = \frac{1}{1+u^2}$$

$$\frac{1-u^2}{1+u^2}$$

$$-\ln|1-u| + C$$

$$-\ln|1-\tan \frac{x}{2}| + C$$

$$1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}$$

$$\frac{2(1-u)}{1+u^2}$$

$$1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}$$

$$\frac{2(1-u)}{1+u^2}$$

$$1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}$$

$$\frac{2(1-u)}{1+u^2}$$

$$1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}$$

$$\frac{2(1-u)}{1+u^2}$$

$$1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}$$

$$\frac{2(1-u)}{1+u^2}$$

$$1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}$$

$$\frac{2(1-u)}{1+u^2}$$

$$1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}$$

$$\frac{2(1-u)}{1+u^2}$$

$$1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}$$

$$\frac{2(1-u)}{1+u^2}$$